Chapter Two

Theory of Demand and Supply

2.1 Theory of demand

- ❖ Demand is one of the forces determining prices.
- ❖ The theory of demand is related to the economic activities of consumers-consumption.

Demand refers to various quantities of a commodity or service that a consumer would purchase at a given time in a market at various prices, given other things unchanged (ceteris paribus).

The quantity demanded of a particular commodity depends on the price of that commodity.

Law of demand: *S*tates that, price of a commodity and its quantity demanded are inversely related i.e., as price of a commodity increases (decreases) quantity demanded for that commodity decreases (increases), ceteris paribus.

2.1.1 Demand schedule (table), demand curve and demand function

A demand schedule states the relationship between price and quantity demanded in a table form.

Table 2.1 Individual household demand for orange per week

Combinations	A	B	C	D	E
Price per kg	5	4	3	2	1
Quantity demand/week	5	7	9	11	13

Demand curve is a graphical representation of the relationship between different quantities of a commodity demanded by an individual at different prices per time period.

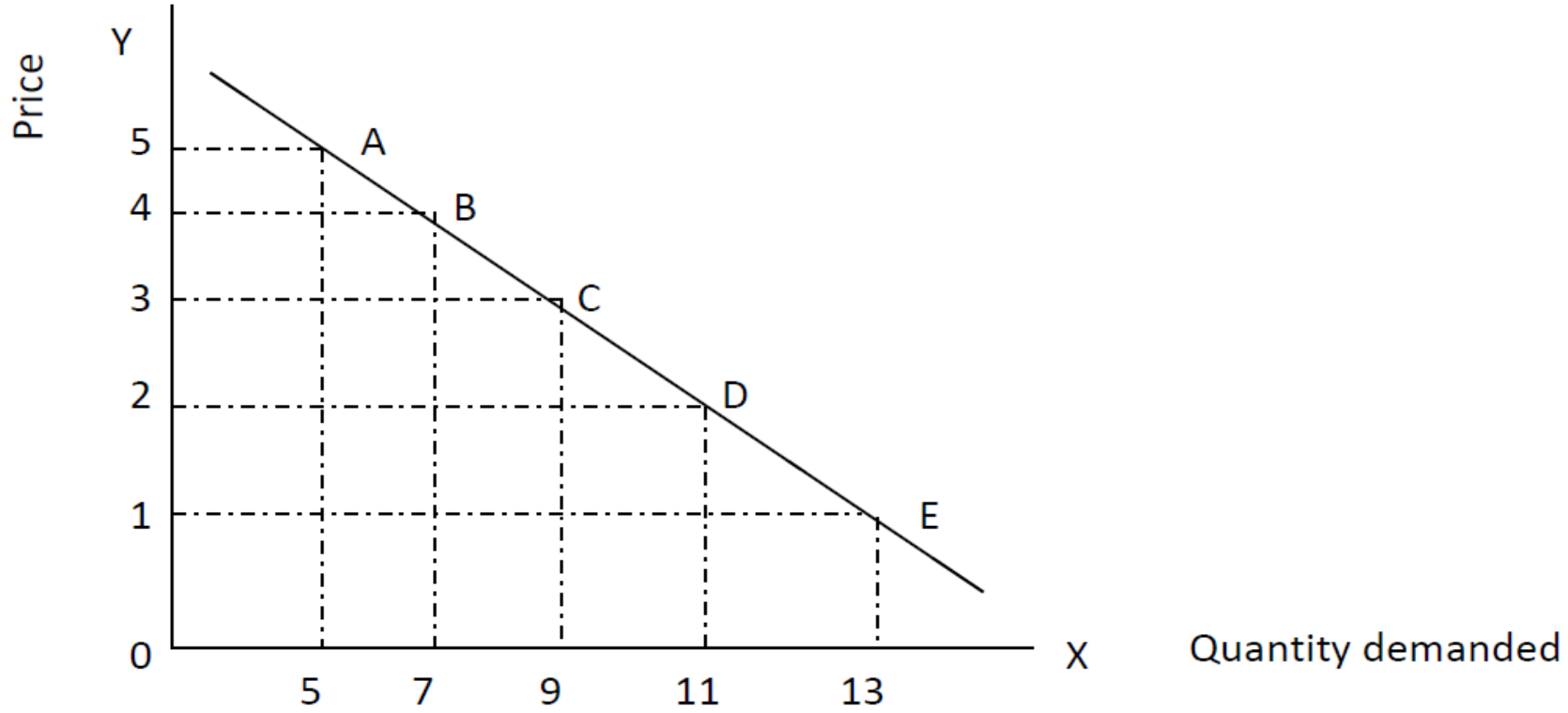


Figure 2.1: Individual demand curve

Demand function is a mathematical relationship between price and quantity demanded, all other things remaining the same.

- A typical demand function is given by: Qd=f(P) where Qd is quantity demanded and P is price of the commodity, in our case price of orange.

Example: Let the demand function be Q = a+ bop

$$b = \frac{\Delta Q}{\Delta P}$$ (e.g. moving from point A to B on figure 2.1 above)

$$b = \frac{7-5}{4-5} = -2$$, where b is the slope of the demand curve

Q = a-2P, to find a, substitute price either at point A or B.

7= a-2(4), a = 15

Therefore, Q=15-2P is the demand function for orange in the above numerical example.

Market Demand: The market demand schedule, curve or function is derived by horizontally adding the quantity demanded for the product by all buyers at each price.

Table 2.2: Individual and market demand for a commodity

Price	Individual demand			Market demand
	Consumer-1	Consumer-2	Consumer-3	
8	0	0	0	0
5	3	5	1	9
3	5	7	2	14
0	7	9	4	20

The following graph depicts market demand curve at price equal to 3



Figure 2.2: Individual and Market demand curve

Numerical Example: Suppose the individual demand function of a product is given by: $P=10 - Q/2$ and there are about 100 identical buyers in the market. Then the market demand function is given by: $P= 10 - Q/2 \leftrightarrow Q/2 =10-P \leftrightarrow Q= 20 - 2P$ and $Qm = (20 – 2P)\ 100 = 2000-200P$

2.1.2 Determinants of demand

The demand for a product is influenced by many factors. Some of these factors are:

I. Price of the product

II. Taste or preference of consumers

III. Income of the consumers

IV. Price of related goods

V. Consumers expectation of income and price

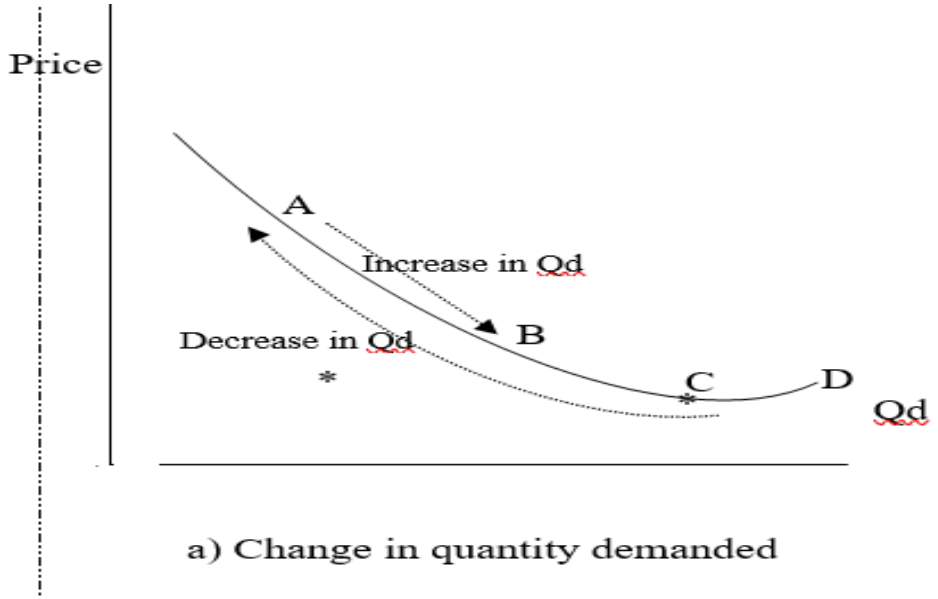
VI. Number of buyers in the market

❖ A change in any of the above listed factors except the price of the good will **change the demand**, while a change in the price, other factors remain constant will bring **change in quantity demanded**.

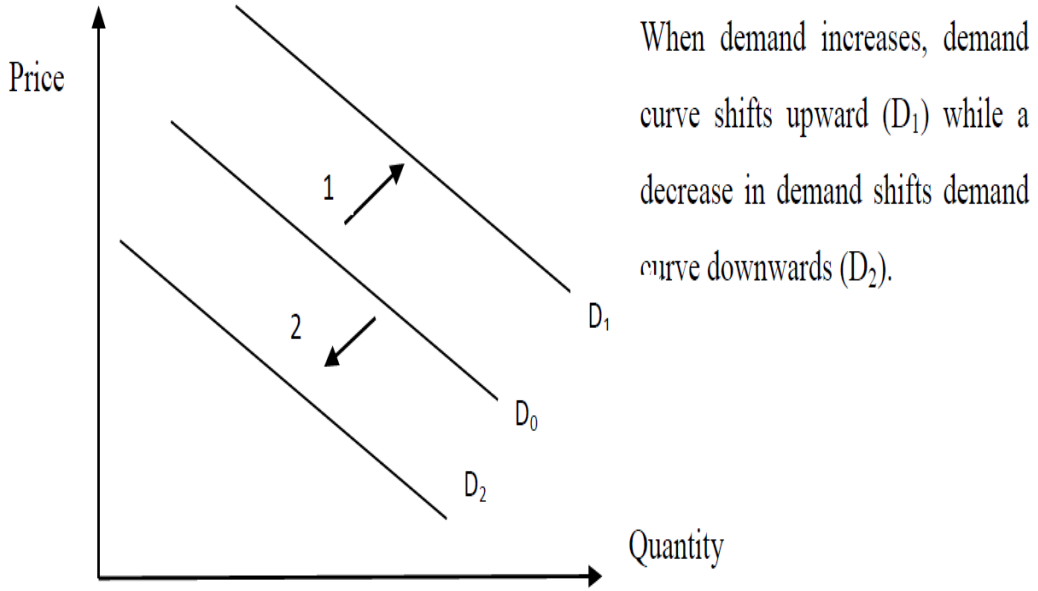
change in quantity demanded: it is caused by change in price. In this case there is movement along the demand curve.

change the demand: it is caused by change in any of the above listed factors except the price.

❖ A change in demand will shift the demand curve from its original location.



a) Change in quantity demanded



When demand increases, demand curve shifts upward (D_1) while a decrease in demand shifts demand curve downwards (D_2).

Figure 2.3: Shift in demand curve

Now let us examine how each factor affect demand.

I. Taste or preference

When the taste of a consumer changes in favor of a good, her/his demand will increase and

the opposite is true.

II. Income of the consumer

Goods are classified into two categories depending on how a change in income affects their demand. These are normal goods and inferior goods. ***Normal Goods*** are goods whose demand increases as income increase, while ***inferior goods*** are those whose demand is inversely related with income.

III. Price of related goods

Two goods are said to be related if a change in the price of one good affect the demand for another good. There are two types of related goods. These are **substitute** and **complimentary goods**.

Substitute goods are goods which satisfy the same desire of the consumer. For example, **tea** and **coffee** or **Pepsi** and **Coca-Cola** are substitute goods.

If two goods are substitutes, then price of one and the demand for the other are directly related. ***Complimentary goods,*** on the other hand, are those goods which are jointly consumed. For example, car and fuel or tea and sugar are considered as compliments. If two goods are complements, then price of one and the demand for the other are inversely related.

IV. Consumer expectation of income and price

Higher price expectation will increase demand while a lower future price expectation will decrease the demand for the good.

V. Number of buyers in the market

Since market demand is the horizontal sum of individual demand, an increase in the number of buyers will increase demand while a decrease in the number of buyers will decrease demand.

2.1.3 Elasticity of demand

Elasticity of demand refers to the degree of responsiveness of quantity demanded of a good to a change in its price, or change in income, or change in prices of related goods.

Commonly, there are three kinds of demand elasticity:

- price elasticity,
- income elasticity, and
- cross elasticity.

i. Price Elasticity of Demand

- Price elasticity of demand means degree of responsiveness of demand to change in price. It indicates how consumers react to changes in price.
- computed as the percentage change in quantity demanded divided by the percentage change in price.

a. Point Price Elasticity of Demand

This is calculated to find elasticity at a given point. The price elasticity of demand can be determined by the following formula.

$$E_d^P = \frac{Percentage\ change\ in\ quantity\ demanded}{percentage\ change\ in\ price} = \frac{\%\Delta Q_d}{\%\Delta P}$$

where $\%\Delta Qd = \frac{Q_1 - Q_0}{Q0} X100$ and

$$\%\Delta P = \frac{P_1 - P_0}{P_0} X100$$

Thus, $E_d^P = \frac{\frac{Q_1 - Q_0}{Q_0} X100}{\frac{P_1 - P_0}{P_0} X100} = \frac{Q_1 - Q_0}{P_1 - P_0} \cdot \frac{P_0}{Q_0} = \frac{\Delta Q}{\Delta P} \cdot \frac{P_0}{Q_0}$

Note that:

- ✓ Elasticity of demand is unit free because it is a ratio of percentage change.
- ✓ Elasticity of demand is usually a negative number because of the law of demand. If the price elasticity of demand is positive the product is inferior.

i) If $|\varepsilon| > 1$, demand is said to be elastic and the product is luxury product

ii) If $0 \leq |\varepsilon| < 1$, demand is inelastic and the product is necessity

iii) If $|\varepsilon| = 1$, demand is unitary elastic.

iv) If $|\varepsilon| = 0$, demand is said to be perfectly inelastic.

v) If $|\varepsilon| = \infty$, demand is said to be perfectly elastic.

Determinants of price Elasticity of Demand

The following factors make price elasticity of demand elastic or inelastic other than changes in the price of the product.

i) **The availability of substitutes:** the more substitutes available for a product, the more

elastic will be the price elasticity of demand.

ii) **Time:** In the long- run, price elasticity of demand tends to be elastic. Because: More substitute goods could be produced. People tend to adjust their consumption pattern.

iii) **The proportion of income consumers spend for a product: -**the smaller the proportion of income spent for a good, the less price elastic will be.

iv) **The importance of the commodity in the consumers' budget:**

Luxury goods: tend to be more **elastic,** example: gold.

Necessity goods: tend to be **less elastic** example: Salt.

ii. Income Elasticity of Demand

It is a measure of responsiveness of demand to change in income.

$$\varepsilon_d^I = \frac{\%\Delta Qd}{\%\Delta I} = \frac{\Delta Q}{\Delta I} \cdot \frac{I}{Q}$$

Point income elasticity of demand:

i) If $\varepsilon_d^I > 1$, the good is **luxury** good.

ii) If $\varepsilon_d^I < 1$ (and positive), the good is **necessity** good,

iii) If $\varepsilon_d^I < 0$, (negative), the good is **inferior** good.

iii. Cross price Elasticity of Demand

Measures how much the demand for a product is affected by a change in price of another good.

$$\varepsilon_{xy} = \frac{\%\Delta Qx}{\%\Delta P_y} = \frac{Q_{x_1} - Q_{x_o}}{P_{y_1} - P_{y_0}} \cdot \frac{P_{y_0}}{Q_{x_0}}$$

i) The cross – price elasticity of demand for substitute goods is **positive.**

ii) The cross – price elasticity of demand for complementary goods is **negative.**

iii) The cross – price elasticity of demand for unrelated goods is **zero.**

2.2 Theory of supply

Supply indicates various quantities of a product that sellers (producers) are willing and able to provide at different prices in a given period of time, other things remaining unchanged.

The law of supply: states that, ceteris paribus, as price of a product increase, quantity supplied of the product increases, and as price decreases, quantity supplied decreases.

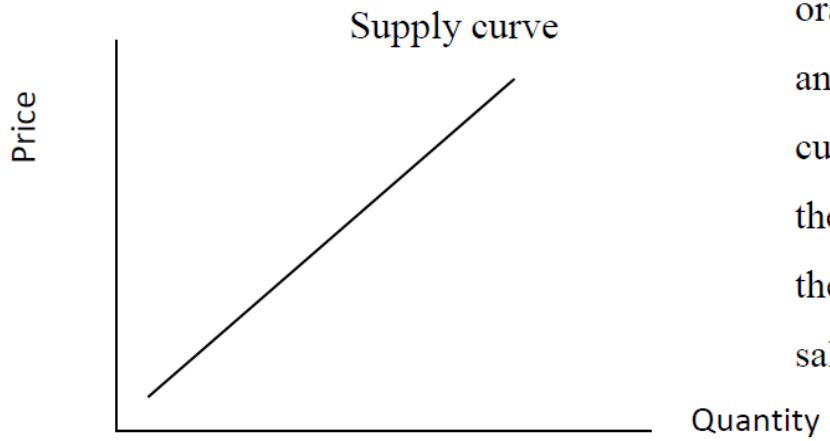
❖ It tells us there is a positive relationship between price and quantity supplied.

2.2.1 Supply schedule, supply curve and supply function

A supply schedule is a tabular statement that states the different quantities of a commodity offered for sale at different prices.

Table 2.3: an individual seller's supply schedule for butter

Price (birr per kg)	30	25	20	15	10
Quantity supplied kg/week	100	90	80	70	60



In this diagram the quantities of oranges are measured along X axis and prices along Y axis. The supply curve slopes upward as we go from the left to the right. This means, as the price rises, more is offered for sale and vice-versa.

Figure 2.5 supply curve

Market supply: It is derived by horizontally adding the quantity supplied of the product by all sellers at each price.

Table 2.4: Derivation of the market supply of good X

Price per unit	Quantity supplied by seller 1	Quantity supplied by seller 2	Quantity supplied by seller 3	Market supply per week
5	11	15	8	34
4	10.5	13	7	30.5
3	8	11.5	5.5	25
2	6	8.5	4	18.5
1	4	6	2	12

2.2.2 Determinants of supply

Apart from the change in price which causes a change in quantity demanded, the supply of a particular product is determined by:

i) price of inputs (cost of inputs)

ii) technology

iii) prices of related goods

iv) sellers ‘expectation of price of the product

v) taxes & subsidies

vi) number of sellers in the market

vii) weather, etc.

2.2.3 Elasticity of supply

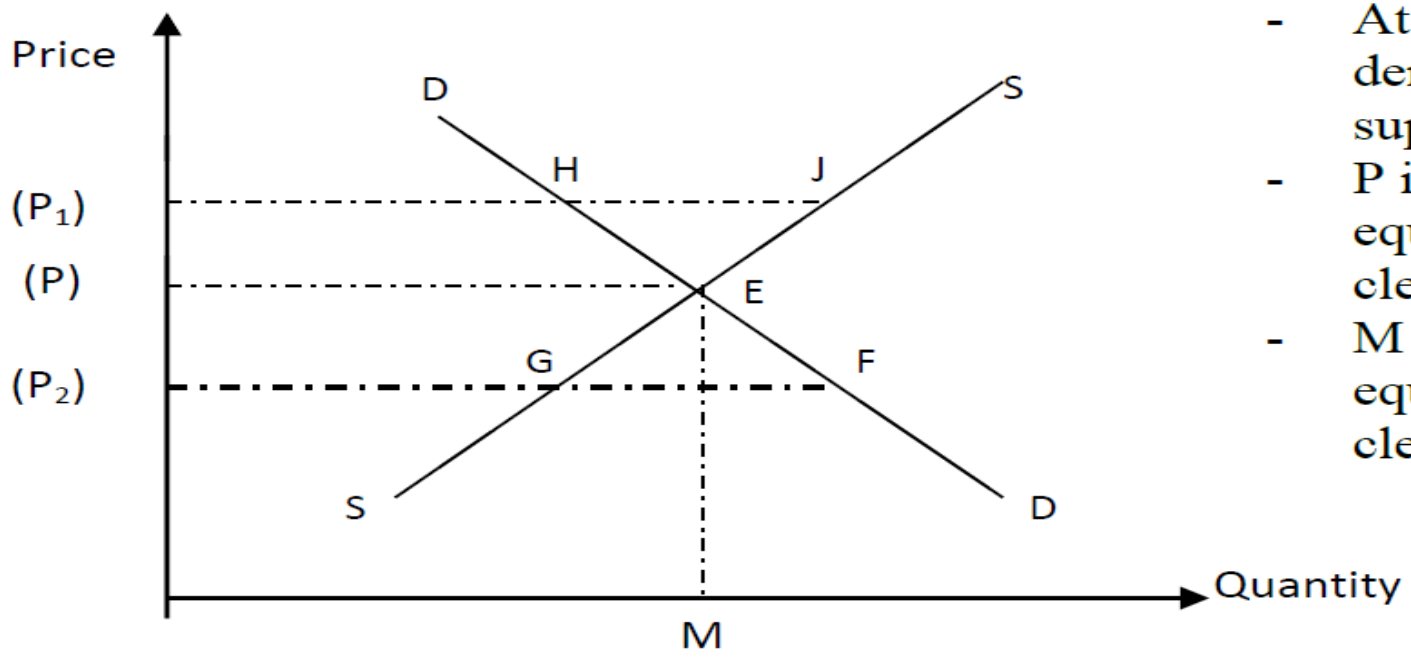
- It is the degree of responsiveness of the supply to change in price.
- It may be defined as the percentage change in quantity supplied divided by the percentage change in price.

$$Es = \frac{\%\ change\ in\ quantity\ \ supplied}{\%\ change\ in\ price} = \frac{\Delta Q}{Q} \div \frac{\Delta P}{P} = \frac{\Delta Q}{\Delta P} * \frac{P}{Q}$$

The interpretations we used for price elasticity of demand also work for price elasticity of supply. That is, if Es>1, supply is elastic; if Es < 1, then supply is inelastic; and if Es = 1, then supply is unitary elastic. In extreme cases, if Es= 0, supply is perfectly inelastic; and if Es = ∞, Then supply is perfectly elastic.

2.3 Market equilibrium

Market equilibrium occurs when market demand equals market supply.



- At point 'E' market demand equals market supply (equilibrium point)
- P is the market equilibrium (market clearing) price.
- M is the market equilibrium (market clearing) quantity.

Figure 2.7: market equilibrium

In the above graph, any price greater than P will lead to market surplus HJ (excess supply). any price less than P will lead to leading to shortage in the market which is equal to GF (excess demand).

Numerical example: Given market demand: $Q_d = 100-2P$, and market supply: $P = (Q_s/2) + 10$

a) Calculate the market equilibrium price and quantity

b) Determine, whether there is surplus or shortage at $P = 25$ and $P = 35$.

Effects of shift in demand and supply on equilibrium

i) when demand changes and supply remain constant

Factors such as changes in income, tastes, and prices of related goods will lead to a change in demand.

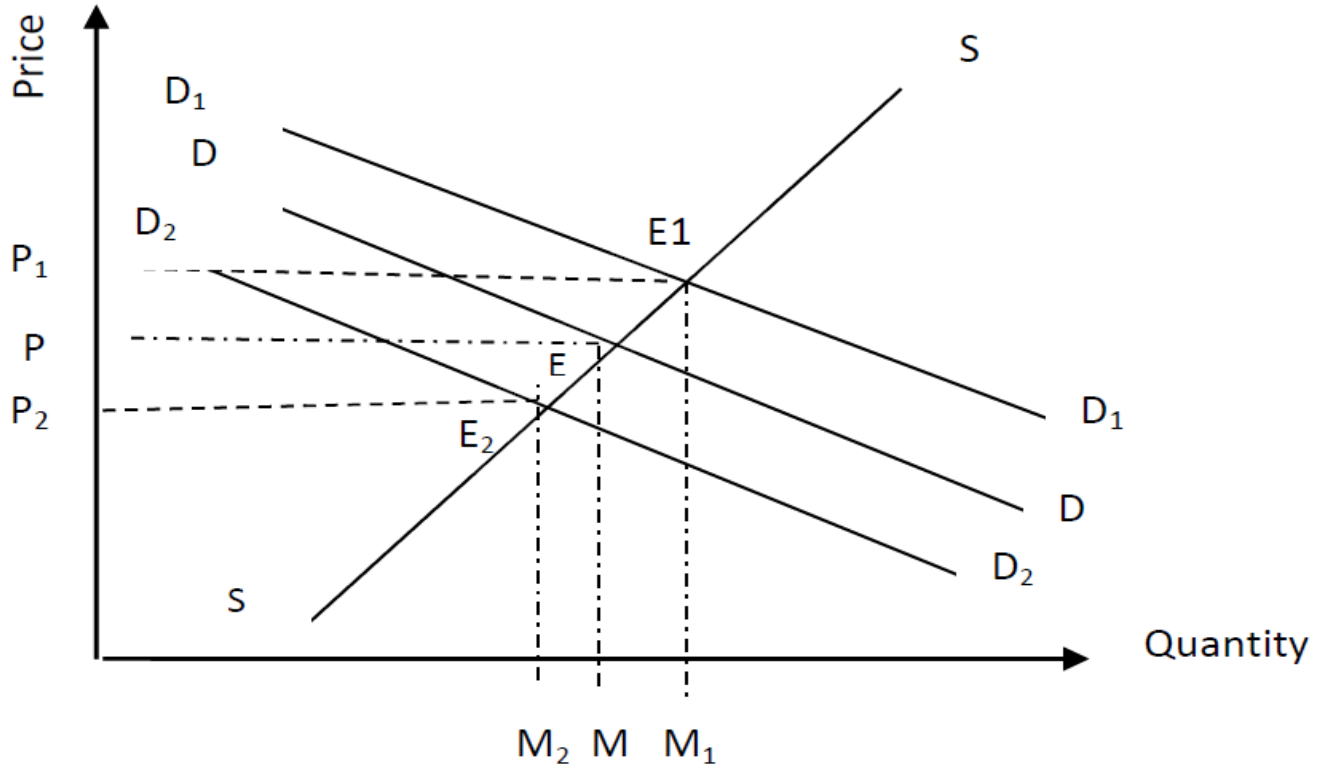


Figure 2.8: The effect of change in demand on market equilibrium

ii. When supply changes and demand remain constant

Changes in supply are brought by changes in technical knowledge and factor prices. The following graph explains the effects of changes in supply.

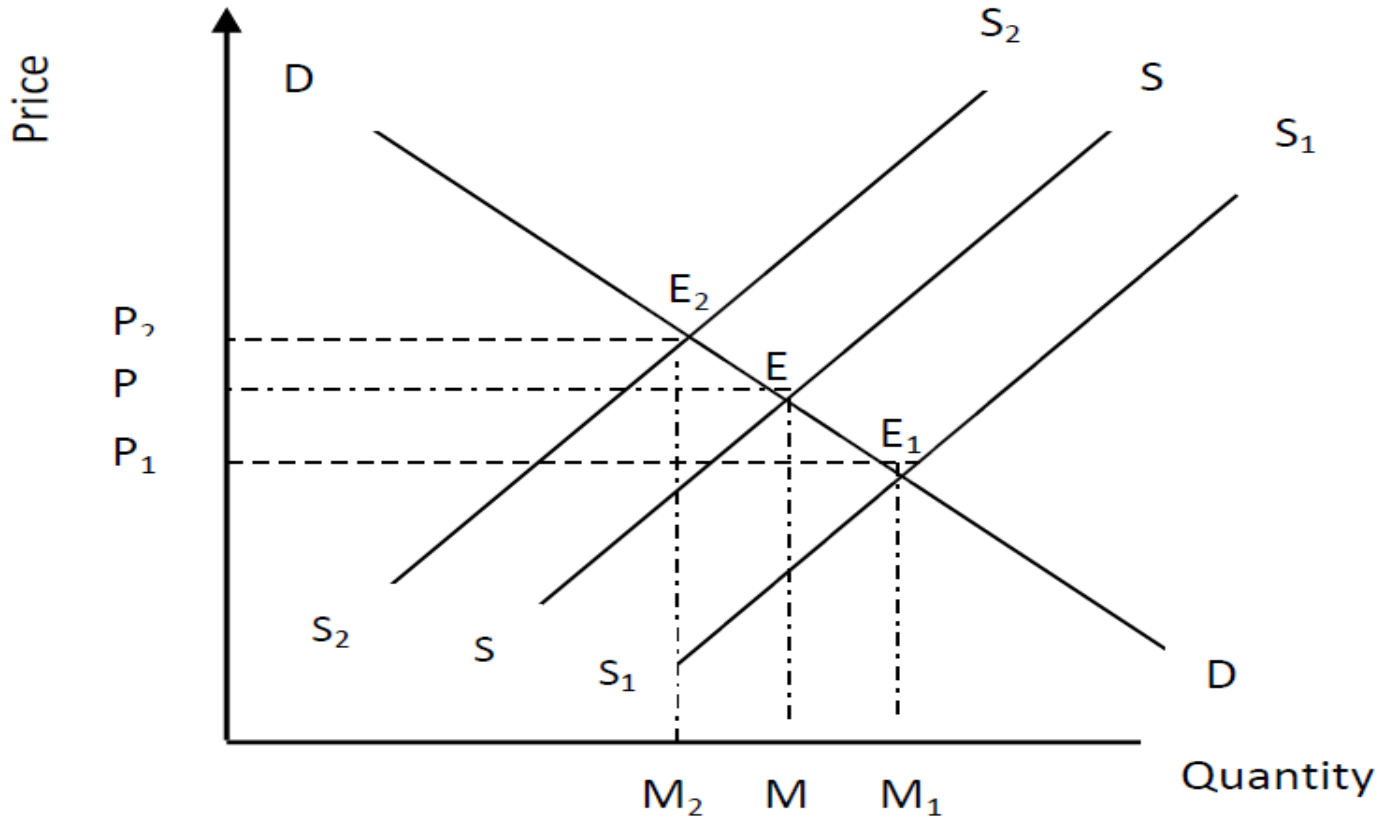


Figure 2.9: The effect of change in supply on market equilibrium